

VIBHAV INNA KEDEGE

MSc Embedded Systems (Honours), BSc Electrical Engineering

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📍 Rijswijk, The Netherlands

• Nationality: Dutch (Netherlands, EU) & Indian (OCI)



PROFESSIONAL EXPERIENCE

Alfen N.V.

DevOps Software Engineer | R&D Department

📅 Oct 2023-Present 📍 Almere, The Netherlands

- Engineering a Distributed Test Automation Platform using **Python, Jenkins, Docker and Azure** that is accelerating firmware delivery cycles by **6x** (from 12 weeks to 2 weeks).
- Pioneered a Real-time Analytics Engine using **FastAPI & PostgreSQL** to automate **5,000+** daily functionality tests analysis, that has enabled multiple teams identify firmware defects **10 weeks earlier**, reclaiming **€150K+** in engineering capacity every quarter.
- Designing enterprise **MLOps** systems for Predictive Maintenance and **LLM-based** Log Analysis to automate root-cause identification and de-risk technical validation for IoT battery and EV charging systems.

KPN (via Capgemini)

Data Engineer (Big Data) | KPN In-Home Analytics

📅 Oct 2021-Sep 2023 📍 The Hague, The Netherlands

- Engineered Scalable Data Pipelines using **Spark and Scala** to ingest and monitor telemetry from **1M+ IoT devices** nationwide, processing billions of daily records for real-time network health analysis.
- Developed a Predictive ML System using **Spark MLlib (Random Forest)** to analyze real-time health metrics. This enabled proactive network management and significantly reduced customer churn.
- Optimized Data Visualization Architectures in **Tableau and Python** to provide executive-level insights into internet quality, transforming raw device logs into actionable business intelligence.

Almende B.V.

Robotics R&D Engineer | R&D Department

📅 Jun 2020-Aug 2021 📍 Rotterdam, The Netherlands

- Developed Distributed Planning Algorithms for **multi-agent SLAM** using **Python, C++, and ROS** that validated agent coordination via simulations for autonomous robot deployments. (Link)
- Published Peer-Reviewed Research on **multi-agent path planning** that translated complex algorithmic findings into a technical case study presented at (Link)

Alstom Transport

System Engineer - Train Networks | Train Control Department

📅 Jul 2017-Jul 2019 📍 Bangalore, India and Sydney, Australia

- Designed and Validated Ethernet-based **Autonomous Train Control Systems**, achieving a **90% pass rate** and completing field validation in India and Australia **3 months before schedule**.

PERSONAL STATEMENT

A proactive Software Engineer with 7+ years of experience specializing in **Data Platforms and MLOps**. Proven track record of designing distributed systems for **1M+ devices** and accelerating software delivery pipelines by **6x** through scalable infrastructure engineering. '

EDUCATION

Delft University of Technology

MSc Embedded Systems (Honours)

📅 2019 - 2021 📍 Delft, Netherlands

- Specialisation - Artificial Intelligence (AI) and Robotics.
- Thesis - "Multi Robot Exploration and Planning in limited communication environments" (Link).
- Published at ICAART 2022 (Link).

National Institute of Technology

Karnataka (NITK-Surathkal)

Bachelor of Technology, Electrical and Electronics Engineering

📅 2013 - 2017 📍 Mangalore, India

- Specialisation - Embedded Software
- Thesis - "Multiple Interface Automation using Distributed Embedded Devices" (Link)

TECHNICAL SKILLS

Core Languages

Python SQL C++ Scala Java

Data & AI / MLOps

Apache Spark Tensorflow MLflow

Prefect MinIO LLMs RAG

Platform & Cloud

Docker Jenkins Azure FastAPI

Linux

Engineering & Systems

Distributed Systems System Design

HIL Testing CI/CD

SELECTED PROJECTS

ML-Powered Stock Prediction & Monitoring System

MLOps Portfolio Project

📅 Jan 2025 – Present

- Building production-grade **MLOps** pipeline for S&P 500 stock prediction with **Streamlit**, **FastAPI** and **MLflow**, implementing automated model training, drift detection, and retraining workflows for **500+** stocks
- Architected end-to-end ML infrastructure using **Python**, **Docker**, and **PostgreSQL** with experiment tracking (**MLflow**), workflow orchestration (**Prefect**), prediction serving (**FastAPI**), and real-time monitoring dashboard (**Streamlit**) ([Live Demo](#))

Local RAG & LLM Orchestration Engine

Technical Case Study

📅 Jul 2025

- Developed a RAG pipeline using **LangChain** and **Ollama** for local LLM inference; integrated a Vector Database for context-aware semantic retrieval. ([Technical Article](#))
- Architected context-injection logic to minimize LLM hallucinations, validating a **Retrieval-Augmented Generation (RAG)** approach for secure, private data processing.

MLOps for Image Segmentation

Freelance Project

📅 Apr 2023 - Jul 2023

📍 Delft, Netherlands

- Architected an MLOps Pipeline to automate training and deployment of **YOLOv8** and **SAM** models using **Python**, reducing model iteration cycles from **weeks to daily updates**.
- Engineered Workflows for **100K+ high-resolution images**; optimized inference and deployment for logistics automation, bridging vision research with production-scale industrial needs.

RL Sensitivity & Optimization Analysis

Technical Case Study

📅 Sep 2020

📍 Delft, NL

- Developed and Benchmarked RL Agents (**SARSA**, **Q-Learning**) within OpenAI Gym by performing multi-dimensional sensitivity analysis to identify optimal hyperparameter convergence. ([Technical Article](#))

HONORS & LEADERSHIP

- **Alumni Advisor, ISA TU Delft**: Mentoring international students on navigating the Dutch higher education system and successfully transitioning into the tech industry.
- **Visionary Engineer Award (Alstom Transport)**: Awarded for excellence in project planning and the optimization of system delivery schedules.
- **President, ISTE (NITK Chapter)**: Led a 100+ member technical student society; managed large-scale engineering workshops and symposiums for 1,000+ participants.
- **Community Volunteer**: Organized STEM-focused educational outreach for underprivileged students in Indian government schools to bridge the digital divide.

CERTIFICATIONS

- Advanced Python Development Techniques ([LINK](#))
- Google Advanced Data Analytics Specialization ([LINK](#))
- DeepLearning.AI TensorFlow Developer Specialization ([LINK](#))
- Professional Scrum Master I ([LINK](#))

MANAGERIAL SKILLS

Project Planning

Collaboration

Customer Interaction

Agile Working

Scrum

Mentoring and Training

KNOWN LANGUAGES

English

Dutch (A2)

German (A1)

Hindi

Kannada

INTERESTS

- **Technical Writing**: Author of deep-dive articles on ML, Algorithms and Distributed Systems on **Medium** and **Personal Blog**.
- **Sports**: Active half-marathon runner (10+ sanctioned events)([Strava](#)) and Tennis
- **Creative Pursuits**: (Photography)